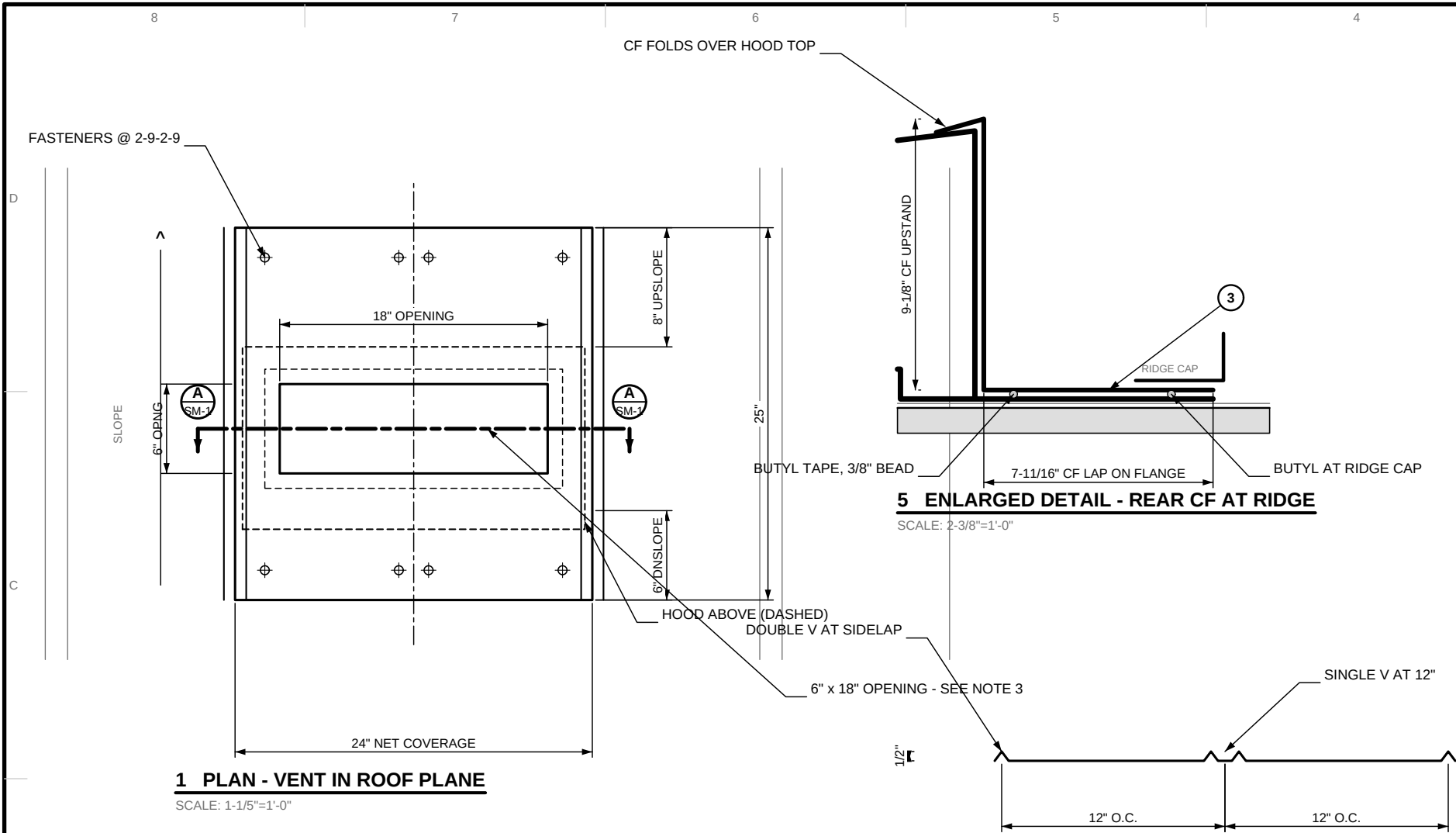
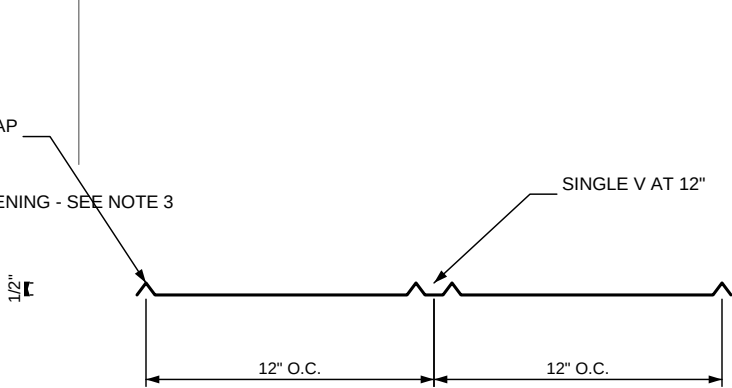




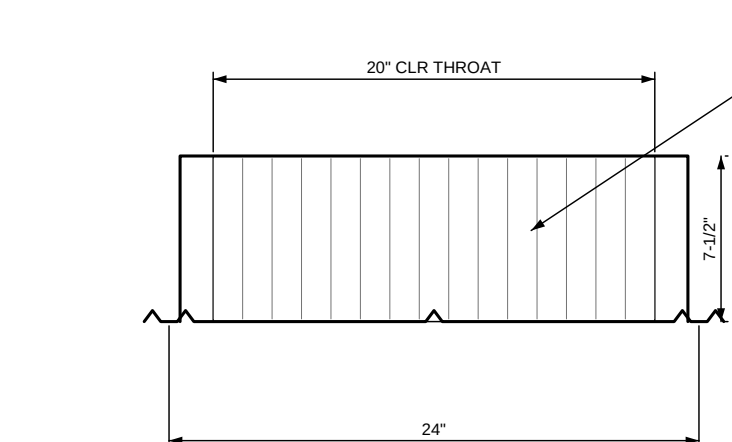
5 ENLARGED DETAIL - REAR CF AT RIDGE

SCALE: 2-3/8"=1'-0"



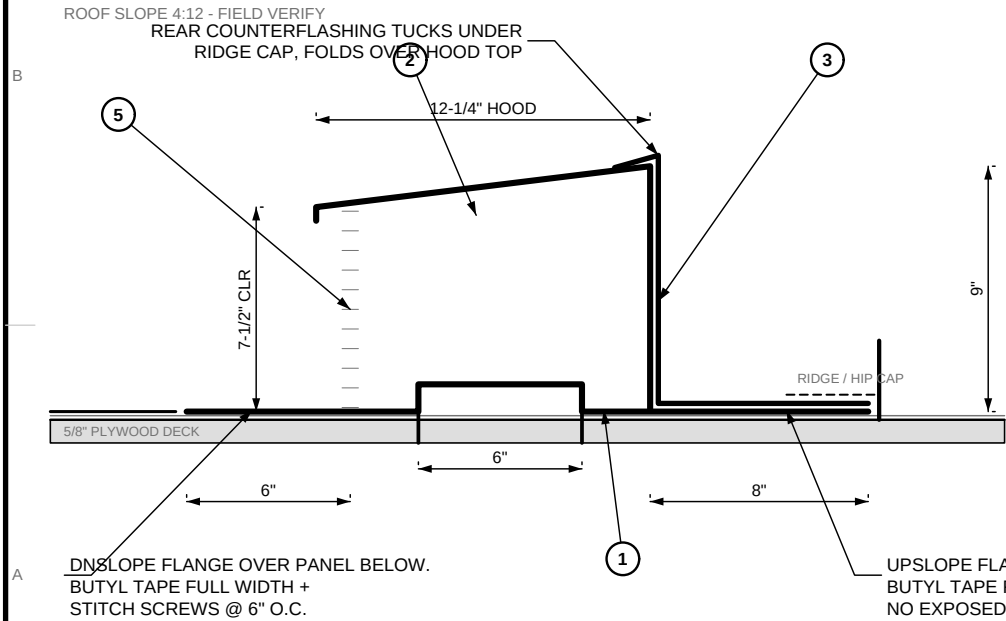
4 METAL SALES 5V-CRIMP - SUBSTRATE PROFILE

SCALE: 3/4"=1'-0"



3 DOWNSLOPE ELEVATION - THROAT

SCALE: 3/4"=1'-0"



2 SECTION A-A - THROUGH VENT CENTRELINE

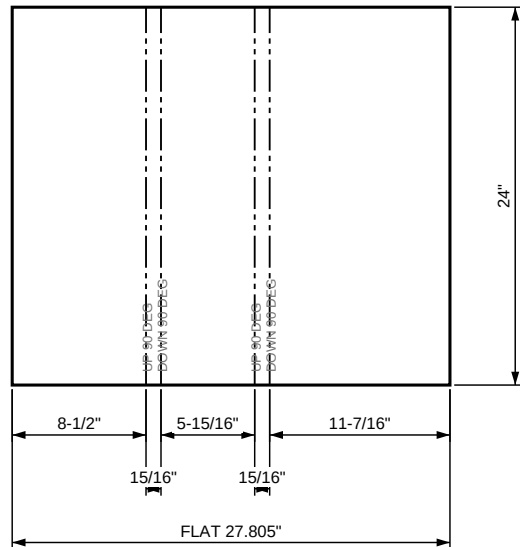
SCALE: 1-3/4"=1'-0"

BILL OF MATERIALS - PER VENT (2 REQUIRED)				
#	DESCRIPTION	MATERIAL	QTY	DTL
1	Base pan / deck flashing, 5V edges	24 ga Galvalume	1	SM-2
2	Hood - top + folded sides	24 ga Galvalume	1	SM-2
3	Rear counterflashing	24 ga Galvalume	1	SM-2
4	Side counterflashing	24 ga Galvalume	2	SM-2
5	Insect screen, 1/4" mesh	Aluminium	1	-
6	Butyl tape sealant, 3/8" bead	ASTM C1311	A/R	SM-2
7	Panel fastener, #9-15 x 1" HH	w/ neoprene wshr	A/R	SM-2
8	Stitch screw, #10-12 x 1" pancake	Cl.4 w/ EPDM	A/R	SM-2

NET FREE AREA - PER VENT	
VENTILATION	VALUE
REQUIRED by roof plan	NFA MIN. 105 S.I.
Deck opening (roof plan)	6" x 18" = 108 sq in
Gross throat opening	20" x 7.5" = 150 sq in
Screen / entry derate	x 0.75 (calculated)
NFA PROVIDED	112 sq in
Check	112 >= 105 OK, margin +8 sq in
Basis	Calculated. Tested figure governs if issued

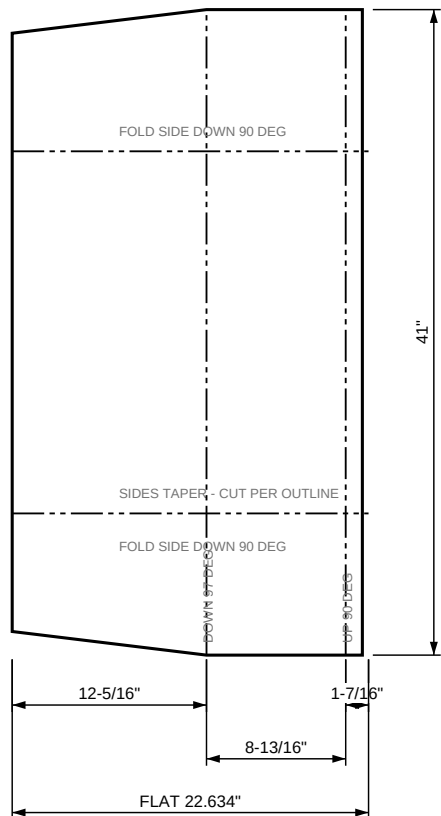
GENERAL NOTES	
1.	FIELD VERIFY all dimensions, roof slope and panel module at both vent locations before fabrication. Dimensions are shop dimensions taken in the plane of the roof and are OUTSIDE unless noted otherwise.
2.	Roof plan requires NFA MIN. 105 S.I. per vent, TYP. OF 2, AT HIGH. Throat is sized to meet it - see the net free area calculation. DCSM publishes no tested NFA.
3.	Deck opening 6" x 18" thru metal roof, insulation and plywood, BETWEEN RAFTERS, per the roof plan. Do not cut a rafter; confirm spacing and direction before cutting.
4.	All components 24 ga Galvalume, ASTM A792 AZ50/AZ55, finish to match the Metal Sales 5V-Crimp roof panel (code 41, non-painted).
5.	Rear counterflashing tucks UNDER the ridge or hip cap and OVER the pan's upslope flange. Both sides counterflash.
6.	Panels run continuous eave to ridge on this roof (up to 39'-3"), so there is no course above to lap under. Vents are located AT HIGH per the roof plan; carry the rear counterflashing under the ridge cap. Downslope flange laps OVER the panel below.
7.	DO NOT SOLDER GALVALUME. Joints to be mechanically locked or riveted and sealed. Isolate from copper and from treated lumber.
8.	Locate each vent within one panel module; do not fasten through a crimp. Coordinate with the panel layout, submittal Tab 12.
9.	Fabricate and install per SMACNA Architectural Sheet Metal Manual, 7th Edition (ANSI/SMACNA 1120-2012) and the project specification.

NO.	DATE	REVISION / RESUBMITTAL
0	09/04/2026	Issued for approval
R&B ROOFING & REMODELING SHOP DRAWING		
Louisville, KY FOR GC / ARCHITECT APPROVAL		
PROJECT		
TEXAS ROADHOUSE — CLARKSVILLE		
LOCATION		
4615 Medical Plaza Way, Clarksville, IN 47129		
SHEET TITLE		
DCSM ROOF VENT - ASSEMBLY		
GENERAL CONTRACTOR		ARCHITECT
Buffalo Construction, Inc.		SEE CONTRACT DOCS
DRAWN BY	CHECKED	DATE
J. ALLEE	-	09/04/2026
SCALE	SUBMITTAL	SHEET
AS NOTED	#2.05 R0 (07 72 00)	SM-1 OF 2



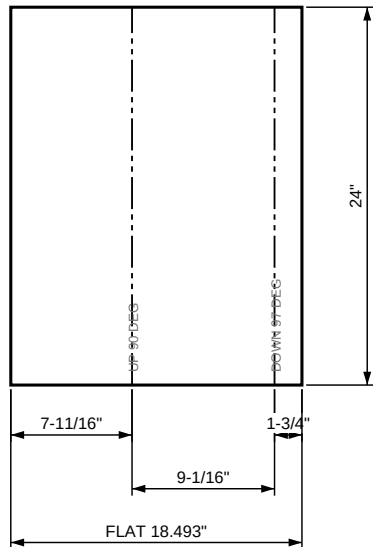
1 FLAT PATTERN - PART 1 BASE PAN

SCALE: GIRTH 28.000" / BLANK 27.805" x 24.000" / QTY 1 EA



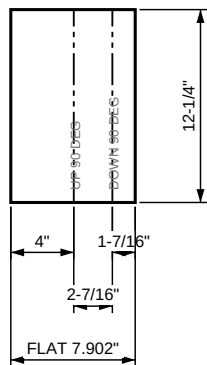
2 FLAT PATTERN - PART 2 HOOD

SCALE: GIRTH 22.741" / BLANK 22.634" x 41.000" / QTY 1 EA



3 FLAT PATTERN - PART 3 REAR CF

SCALE: GIRTH 18.600" / BLANK 18.493" x 24.000" / QTY 1 EA



4 FLAT PATTERN - PART 4 SIDE CF

SCALE: GIRTH 8.000" / BLANK 7.902" x 12.250" / QTY 2 EA

BEND SCHEDULE							
PART	BEND	ANGLE	DIR	INSIDE R	BEND ALLOW	BEND DED	NOTE
1	1	90.0	UP	0.028"	0.062"	0.049"	dnslope curb
1	2	90.0	DOWN	0.028"	0.062"	0.049"	curb to pan
1	3	90.0	UP	0.028"	0.062"	0.049"	upslope curb
1	4	90.0	DOWN	0.028"	0.062"	0.049"	curb to flange
2	1	97.0	DOWN	0.028"	0.066"	0.058"	top to back plate
2	2	90.0	UP	0.028"	0.062"	0.049"	back plate to flange
3	1	90.0	UP	0.028"	0.062"	0.049"	lap to upstand
3	2	97.0	DOWN	0.028"	0.066"	0.058"	upstand to fold
4	1	90.0	UP	0.028"	0.062"	0.049"	lap to upstand
4	2	90.0	DOWN	0.028"	0.062"	0.049"	upstand to return

CUT LIST - PER VENT x 2 VENTS					
PART	MATERIAL	QTY	GIRTH	FLAT	BLANK L x W
1 BASE PAN	24ga galv	1 x 2	28.000"	27.805"	27.805" x 24.000"
2 HOOD	24ga galv	1 x 2	22.741"	22.634"	22.634" x 41.000"
3 REAR CF	24ga galv	1 x 2	18.600"	18.493"	18.493" x 24.000"
4 SIDE CF	24ga galv	2 x 2	8.000"	7.902"	7.902" x 12.250"

FASTENER & SEALANT SCHEDULE		
LOCATION	FASTENER / SEALANT	NOTE
Panel to wood deck	#9-15 x 1" Type A hex head	5/8" OD stl washer + bonded neoprene
spacing	2-9-2-9 across the 24" module	3'-0" o.c. up the slope
Field / panel ends	#10-14 x 1" wood screw	per Metal Sales P5V-9
Flashings & closures	#10-12 x 1" pancake head	12" o.c.
Sidelap, pitch < 3:12	SM7108 sealant at each panel	per roof plan note
Concealed laps	Butyl tape, 3/8" bead	ASTM C1311, non-curing
Exposed joints	Polyurethane, gun grade	ASTM C920, tooled
Embedment	Wood screws min 3/4" into deck	UFGS 07 60 00

OPEN ITEMS - RESOLVE BEFORE FABRICATION	
1.	SUBMITTAL DEFECT: the Tab 9 cut sheet is titled "Proper Installation of a Vent on a STANDING SEAM Metal Roof System" and states the vent "sits on TOP of the standing seam rather than on the flat section of the panel." The TOC and tab divider call it 5V. No 5V vent literature was submitted and nothing describes how it laps into a 5V panel. The roof plan calls the panel "5 RIB V-GROOVE".
2.	DCSM publishes NO dimensions and NO tested net free area for this vent. The throat is sized here to MEET the roof plan's NFA MIN. 105 S.I.; the calculation is shown on SM-1. A tested figure from the manufacturer governs and is requested.
3.	No metal-roof penetration detail was submitted. Metal Sales manual p.17 cross-references "ROOF PENETRATION (SEE PAGE 25)" but only manual pages 17-19 are in the binder.
4.	Roof slope at each vent: the roof plan notes sidelap sealant is required on pitches LESS THAN 3:12, but Metal Sales publishes 3:12 as the panel's MINIMUM slope. Confirm the actual pitch at both vent locations and whether a sub-3:12 plane is involved - that is a manufacturer conflict, not just a sealant note.
5.	Confirm rafter spacing and direction at both vent locations. The 6"x18" opening must fall between rafters without cutting one.
6.	DCSM's own 5V spec sheet states 23-1/2" coverage while their Miami-Dade NOA 19-0109.04 drawing shows 24". The specified Metal Sales panel is 24". Confirm the vent is built to the 24" module.
7.	Sidelap double-V spread, V rib base width and V included angle are not published by Metal Sales, DCSM or the NOA. Measure from an actual panel.
8.	Fastener conflict within the binder: p.35 lists #10-14 x 1" ABMP XL, UL 580 Const. #435; p.58 cites Const. #579 & #453, which specify #9-15 x 1" Type A. Confirm which governs.
9.	Vent gauge: 26 ga to match the roof panel per DCSM's "same gauge as your main roof" rule, or 24 ga per UFGS Table I for base flashing. This drawing shows 24 ga.
10.	Roof plan notes no direct contact between the leak barrier and the roof membrane (not compatible, see 6/A8). Confirm the underlayment used at the vent does not conflict where the metal roof meets the membrane.

FORMING NOTES	
1.	24 ga Galvalume, t = 0.0276" coated (0.0239" base steel, ASTM A792). Minimum inside bend radius 1t. Confirm against the mill cert.
2.	Flats developed with K = 0.42: BA = $\frac{1}{\pi(180)}(A)(R + K t)$; BD = $2 \tan(A/2)(R + t)$ - BA. FLAT = GIRTH - sum of deductions.
3.	K-factor varies with tooling, temper and grain. Bend a coupon and measure before running production coil.
4.	Phantom lines are FOLD lines - do not cut.
5.	Hood sides taper from the throat height at the drip edge to the full rise at the back. Cut the blank to the outline shown, then fold.
6.	Hem all exposed edges 1/2". Corners lapped and riveted, then sealed.
7.	DO NOT SOLDER GALVALUME - the Al-Zn coating will not tin.

NO.	DATE	REVISION / RESUBMITTAL
0	09/04/2026	Issued for approval

R&B ROOFING & REMODELING SHOP DRAWING

Louisville, KY

PROJECT
TEXAS ROADHOUSE — CLARKSVILLE

LOCATION
4615 Medical Plaza Way, Clarksville, IN 47129

SHEET TITLE
DCSM ROOF VENT - FABRICATION

GENERAL CONTRACTOR Buffalo Construction, Inc.	ARCHITECT SEE CONTRACT DOCS
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DRAWN BY J. ALLEE	CHECKED -	DATE 09/04/2026	JOB NO. 26-TR-009
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SCALE AS NOTED	SUBMITTAL #2.05 R0 (07 72 00)	SHEET SM-2 OF 2
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